UF Hamilton School for Classical and Civic Education
UNIVERSITY of FLORIDA

Karl Marx and His Critics

Capitalism and Its Critics
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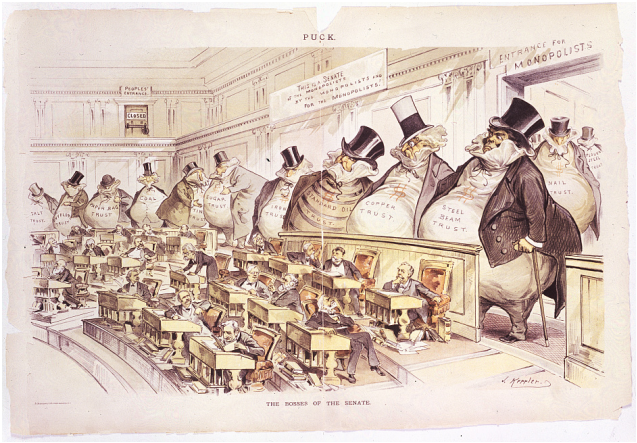
Fall 2026

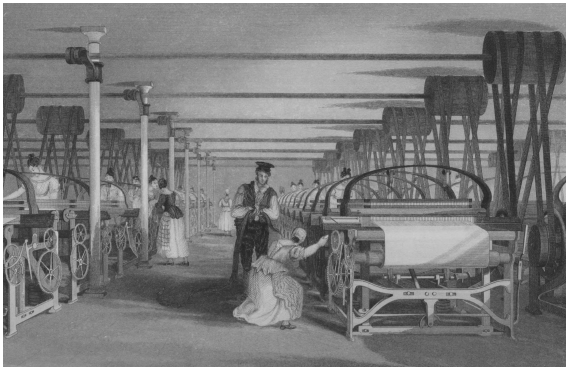
What is capitalism?



Capitalism: a working definition

- Private ownership of the means of production (land, machines, firms).
- Production for sale in markets, coordinated by prices.
- Most people work for wages, hired by the owners of capital.

Britain, 1835–1842



Housing in industrial Birmingham

Birmingham Daily Gazette, 1901

"The air is heavy with a sooty smoke and with acid vapours, and here it is that the poor live and wither away and die. [. . .] Look at the begrimed windows, the broken glass, the apertures stopped with yellow paper or filthy rags; glance in at the rooms where large families eat and sleep every day and every night, amid rags and vermin, within dank and mildewed walls from which the blistered paper is drooping, or the bit of discoloration called 'paint' is peeling away."

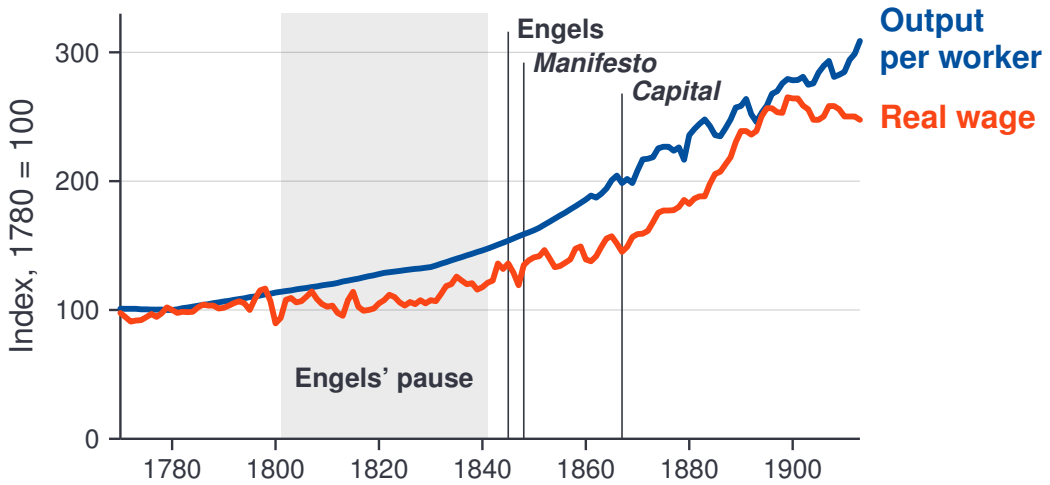
Thomas Street, Birmingham
c. 1882

Children in the coal mines

Sarah Gooder, aged 8, Children's Employment Commission (Mines), 1842

"I'm a trapper in the Gauber Pit. I have to trap without a light, and I'm scared. I go at four and sometimes half-past three in the morning, and come out at five and half-past, I never go to sleep. Sometimes I sing when I've light, but not in the dark; I dare not sing then. I don't like being in the pit."

Were workers getting better or worse off?



Note: Great Britain, output per worker and the average real wage, both 1780 = 100. Allen (2009), Fig. 1.

Engels' pause

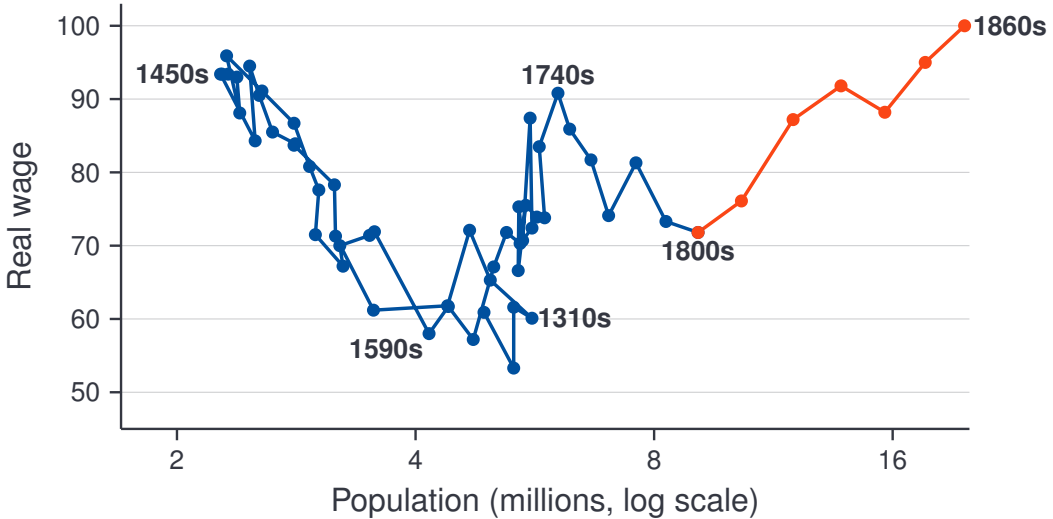
- Real wages barely grew, 1780–1830. GDP per person rose about two-thirds between 1770 and 1860.
- Productivity grew first; capital's share rose from about 20% to over 40%.
  - "Engels' pause": a half-century of delayed gains. Clearer wage catch-up came after 1840.
- Men's annual hours rose from about 2,500 to 3,300, 1760–1830: hourly earnings fared worse than annual earnings.
- Manchester grew from under 20,000 around 1750 to about 330,000 by 1850. Engels lived there in 1842–44.

Allen (2009); Voth (2001); Thomas and Dimsdale (2017); CAMPOP.

Population and wages

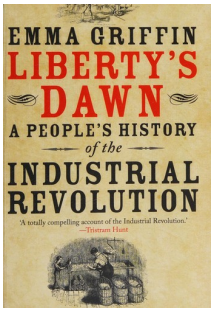
- Before 1760, population could grow about 0.4% a year without lowering real wages; after 1760, about 1.5%.
- Then earlier marriage raised births and fewer epidemics cut deaths: growth peaked at 1.5% a year around 1815.
- Births mattered more: without the rise in births, 1840 real wages would have been about 48% higher; without fewer deaths, about 27%.
- Marx and Engels saw a real shock but mistook a transient phase for a law of motion.
  - By 1900, British real wages were about twice their 1820 level.

Crafts and Mills (2022); Thomas and Dimsdale (2017).



Note: England, decade averages, 1280s–1860s. Real day wage of London building craftsmen, 1860s = 100. Allen (2001), wages; Clark (2005), population.

Liberty's Dawn: workers in their own words



- About 350 working-class autobiographies, 1760–1900, nearly all by men.
- Gains: higher and steadier earnings, literacy, political action, more choice in courtship and marriage.
- Limits: more and younger children at work; women largely shut out; the writers were literate and self-selected.

Griffin (2013).

Moving to the city: pull, push and penalty

- Pull: in the 1830s, money wages in English towns were about 73% above farm wages (Williamson).
- Push: enclosure could remove access to common land. Evidence links parliamentary enclosure to migration toward the industrial North, but not clearly to factory work.
- Penalty: life expectancy at birth was about 25–28 years in Manchester, Liverpool and Glasgow around 1840, against about 40 nationally.

Williamson (1990); Heldring et al. (2024); Davenport (2020); Davenport (2021).

Tocqueville on pauperism

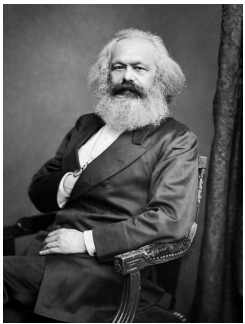
Tocqueville, *Memoir on Pauperism*, 1835

"Cross the English countryside and you will think yourself transported into the Eden of modern civilization [. . .]. Now look more closely at the villages; examine the parish registers, and you will discover with indescribable astonishment that one-sixth of the inhabitants of this flourishing kingdom live at the expense of public charity."

Alexis de Tocqueville
(1805–1859)

A note on Karl Marx, 1818–1883

- Born in Trier, 1818. Doctorate in philosophy, University of Jena, 1841.
- Edited the *Rheinische Zeitung* (1842–43) until Prussia banned it; Paris 1843–45, Brussels 1845–48.
- Wrote the *Communist Manifesto* with Friedrich Engels; published in London, February 1848.
- Lived in London from 1849 to his death in 1883, partly on Engels's money.
- *Capital*, vol. I, 1867. Engels edited vols. II and III from his drafts: 1885, 1894.

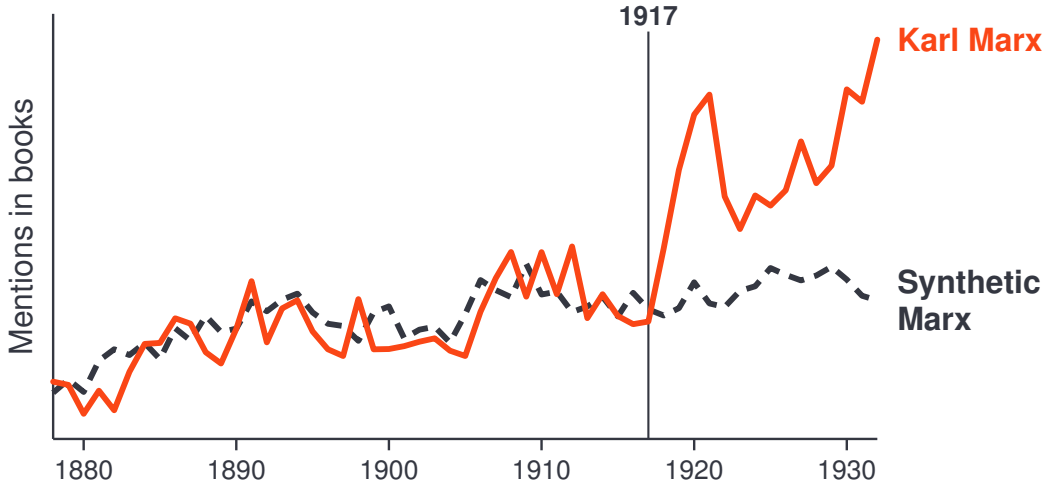


Karl Marx
(1818–1883)

Mark Blaug on Marx

Mark Blaug, *Economic Theory in Retrospect*, 1997

"Marx has been reassessed, revised, refuted and buried a thousand times but he refuses to be relegated to intellectual history."

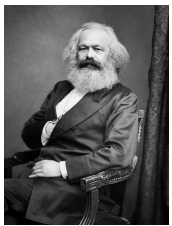


Note: How often “Karl Marx” appears in English-language books (Google Ngram), vs. a “synthetic Marx” built from other authors to track him before 1917. Magness and Makovi (2023), replication data.

Capitalism and class conflict

Marx and Engels, *Manifesto of the Communist Party*, 1848

"Our epoch, the epoch of the bourgeoisie, possesses, however, this distinct feature: it has simplified class antagonisms. Society as a whole is more and more splitting up into two great hostile camps, into two great classes directly facing each other — Bourgeoisie and Proletariat."



Karl Marx
(1818–1883)

Friedrich Engels
(1820–1895)

Utopian vs. scientific socialism

Henri de Saint-Simon
(1760–1825)

Industrialists and scientists, not nobles and priests, should run society.

Charles Fourier
(1772–1837)

Cooperative communities of about 1,600 people, housed in a “phalanstère”; work made attractive.

Robert Owen
(1771–1858)

The model mill village of New Lanark; then New Harmony, Indiana (1825–27).

⇒ Engels: Marx’s two discoveries made socialism “a science”: historical materialism, and the secret of profit (surplus value).

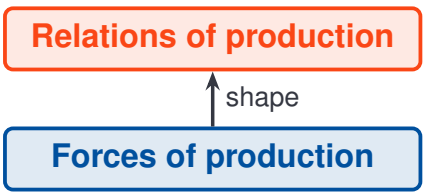
Marx's core claims

- Material conditions: production, property and class drive historical development and shape ideas and consciousness.
- Dialectics: contradictions, especially class conflicts, drive social change.
- Human beings are social and productive. Labor is a means of self-realization.
- Under capitalism, workers are alienated from their product, their activity, fellow humans and their species-being.

Marx (1844); Marx (1859); Marx and Engels (1848).

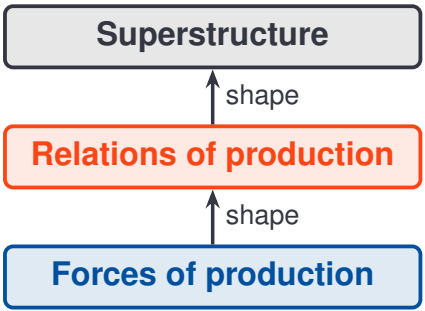
Forces of production Tools, machines, skills

Historical materialism



Who owns what:
lord and serf, capitalist and worker

Tools, machines, skills



Law, politics, religion, ideas

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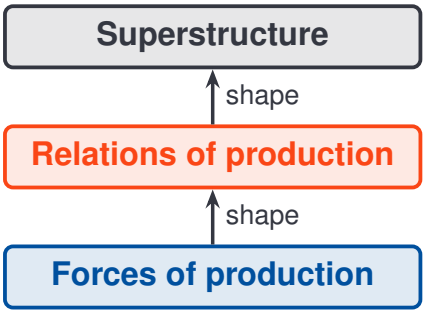

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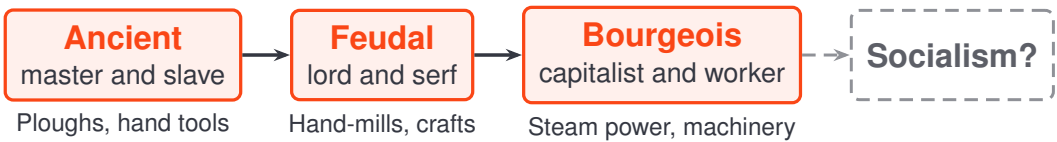


The hand-mill and the steam-mill

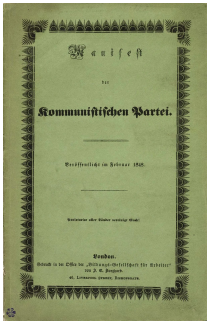
Marx, *The Poverty of Philosophy*, 1847

"The hand-mill gives you society with the feudal lord; the steam-mill, society with the industrial capitalist."

Capitalism's productive power

Marx and Engels, *Manifesto of the Communist Party*, 1848

"The bourgeoisie, during its rule of scarce one hundred years, has created more massive and more colossal productive forces than have all preceding generations together."

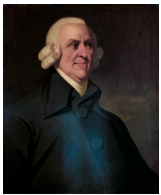

Manifest
der
Kommunistischen Partei.
London.

Manifest der Kommunistischen Partei
London, February 1848

Smith, Ricardo and Marx on value

- The classical questions: what sets relative prices, and how output splits into wages, profits and rents.
- Smith: labor alone sets exchange values only in the "early and rude state" before capital and land are owned: one beaver for two deer.
- Ricardo: relative labor quantities regulate relative values, but goods that "cannot be brought to market in the same time" differ in value.

$\Rightarrow$ Marx: labor is the substance of value, so profit too must be explained by labor.



Adam Smith
(1723–1790)

David Ricardo
(1772–1823)

Labor vs. labor power

- Labor is the work; labor power is the capacity to work.
  - The worker sells labor power by the day.
  - A mill owner who hires a spinner pays for his capacity to work that day, not for the yarn he spins.
- Marx, *Wage Labour and Capital*, ch. 2: “Labour-power, then, is a commodity, no more, no less so than is the sugar.”

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If the source of exchange value is labor,
then where does profit come from?

Marx’s Theory of Capitalism

What do a coat and 20 yards of linen have in common?

Socially necessary labor time

- Marx: two goods that trade as equals “must therefore be equal to a third, which in itself is neither the one nor the other.”

Socially necessary labor time

- Marx: two goods that trade as equals “must therefore be equal to a third, which in itself is neither the one nor the other.”
- The third thing is labor. Value is the labor time needed “under the normal conditions of production, and with the average degree of skill and intensity prevalent at the time.”
- A slow tailor’s coat is not worth more: only socially necessary time counts.

Marx (1867), vol. I, ch. 1.

Selling to buy vs. buying to sell

Commodity (C): a good made to be sold. M: money.

Selling in order to buy

Selling to buy vs. buying to sell

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Labor power

Labor power: the capacity to work, which the worker sells to the capitalist by the day.

- Its value = the labor time needed to produce the worker's means of subsistence, the goods he lives on: in Marx's example, **6** hours a day.
- Marx, *Capital* I: what counts as subsistence has "a historical and moral element" ⇒ not a bare biological minimum.

Free in a double sense

Means of production: the land, tools, machines and materials that work needs.

- The worker is “free in the double sense”: free to sell his labor power, and free of any means of production.
- Marx: capitalism began with “the historical process of divorcing the producer from the means of production”.
- Marx’s name for it: primitive accumulation. Not Smith’s story of saving and specialization.

Marx (1867), vol. I, chs. 6 and 26.

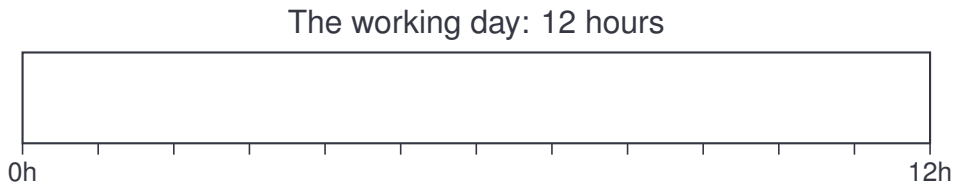
Primitive accumulation

Marx, *Capital*, vol. I, 1867

"And the history of this, their expropriation, is written in the annals of mankind in letters of blood and fire."

Thomas Faed, *The Last of the Clan*, 1865

The working day



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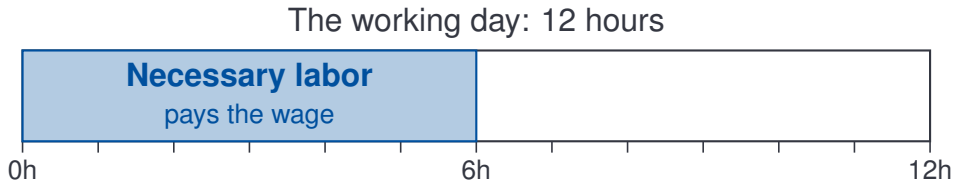

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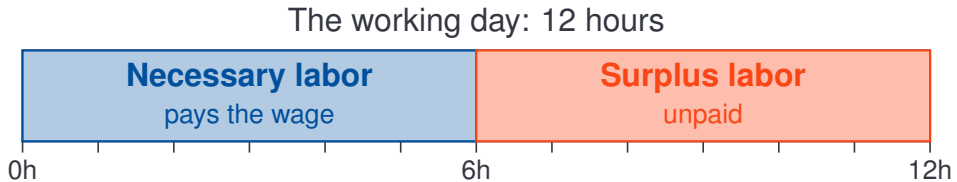
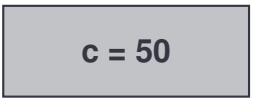


Constant capital, variable capital and surplus value



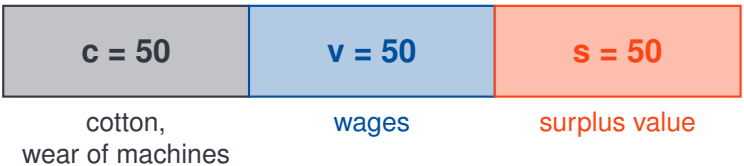
cotton,
wear of machines

- **c**, constant capital: machines and materials pass on their value, unchanged.

Constant capital, variable capital and surplus value

- **c**, constant capital: machines and materials pass on their value, unchanged.
- **v**, variable capital: labor power, says Marx in *Capital* I, “reproduces the equivalent of its own value, and also produces an excess”.

Constant capital, variable capital and surplus value



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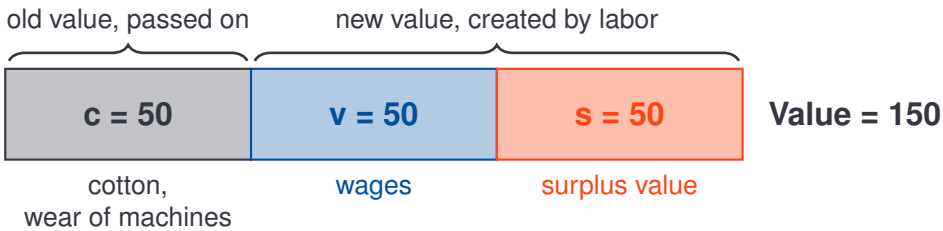

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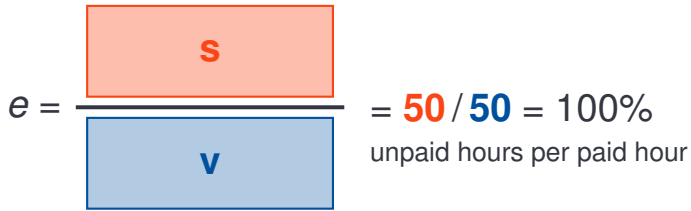
Exploitation, a technical term

Rate of surplus value (e): unpaid hours per paid hour, Marx's measure of exploitation.



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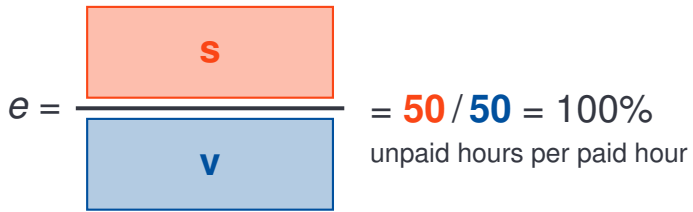
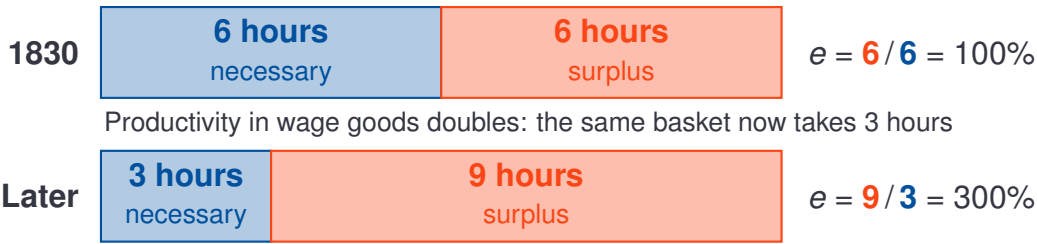


- Mechanism: **6** unpaid hours, yet the wage is the full value of labor power. No fraud.
- Moral charge: Marx calls e "an exact expression for the degree of exploitation of labour-power by capital".
- Testable claim: what Marx predicts for wages, profits and hours as capital grows.

Marx (1867), vol. I, chs. 7–9.

Relative surplus value

Relative surplus value



- Relative surplus value: the necessary part of the same day shrinks from **6** to **3** hours.
- *e* rises from 100% to 300%, while the worker buys the same basket.
- Marx: the price of labor power can keep falling while “the mass of the labourer’s means of subsistence” grows.

Marx (1867), vol. I, chs. 12 and 17.

Dead labor vs. living labor

Marx, *Capital*, vol. I, 1867

"Capital is dead labour, that, vampire-like, only lives by sucking living labour, and lives the more, the more labour it sucks."

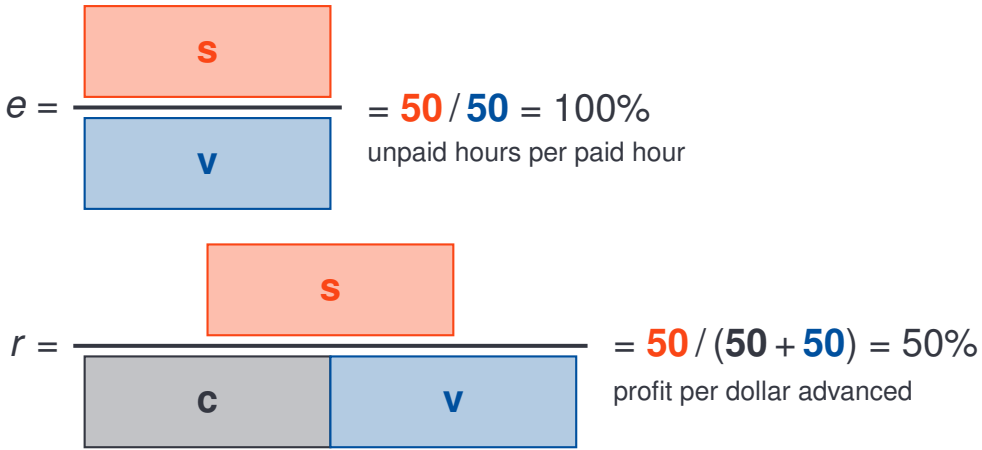
Nosferatu, 1922

The rate of profit

$$e = \frac{s}{v} = 50/50 = 100\%$$

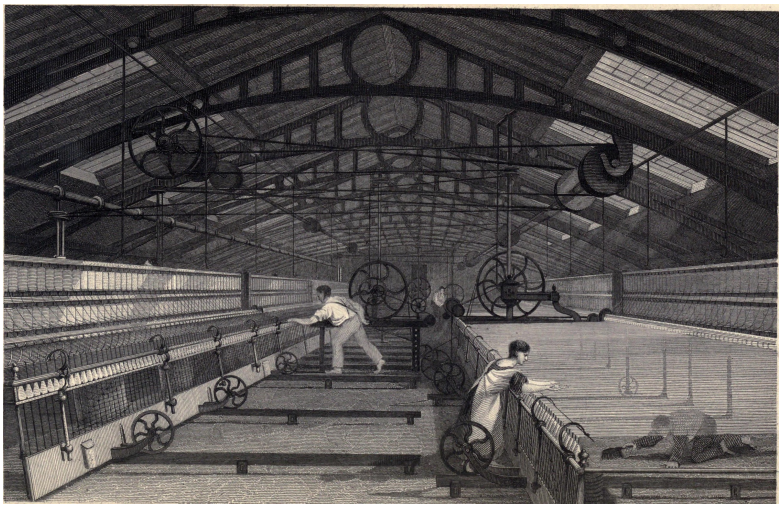
unpaid hours per paid hour

The rate of profit



Same mill, two ratios: **s** against wages (exploitation), **s** against all capital (profit).

Why buy machines?



Machines and extra profit

- The first mill to mechanize makes cloth in less than the socially necessary time and sells near the old value: an “extra surplus-value”.
- Rivals copy. The extra “vanishes, so soon as the new method of production has become general”.
- The mill now has more **c** per worker.

Marx (1867), vol. I, ch. 12.

The falling rate of profit

Mechanization puts more machines and materials (**c**) behind each dollar of wages (**v**).
Marx's composition of capital: $q = c/v$.

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Divide the top and the bottom by v:

$$r = \frac{s}{c+v} = \frac{s/v}{c/v+1} = \frac{e}{1+q}$$

s/v is e, the rate of surplus value, and c/v is q. Hold e fixed: the higher q, the lower r.

The falling rate of profit

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	c	**v**	**s**	***e***	***q***	$r = e/(1+q)$
The 1830 mill	**50**	**50**	**50**	100%	1	$1/(1+1) = 50\%$
The 1860 mill	**80**	**20**	**20**	100%	4	$1/(1+4) = 20\%$

Counteracting influences on the rate of profit

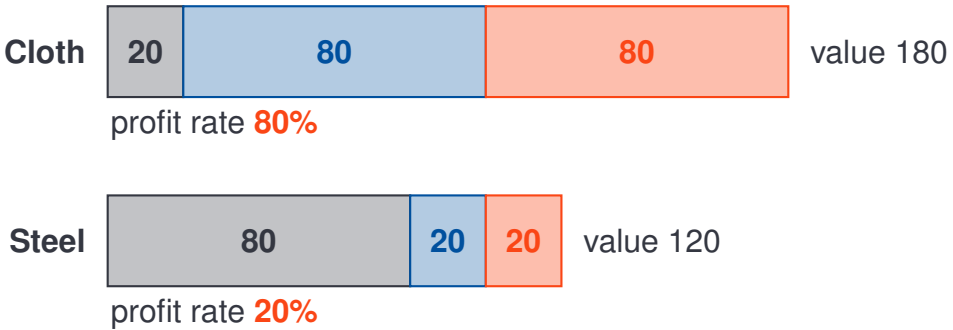
- Marx called the fall a *tendency*, checked by "counteracting influences".

Marx's six counteracting influences	
Longer, harder working days	Wages pushed below their value
Machines and materials get cheaper	Jobless workers keep labor cheap
Cheaper imported inputs and food	Railway companies settle for low dividends

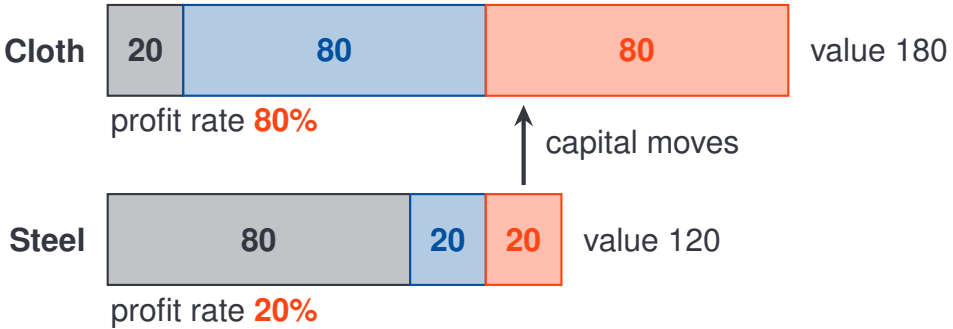
- The formula is arithmetic; the fall is a prediction. If longer, harder days raise e to 250%, the 1860 mill (**80**, **20**, **50**) is back at $r = 2.5/(1 + 4) = 50\%$.

Marx (1894), vol. III, chs. 3 and 13–14.

Cloth vs. steel



Equalizing profit rates



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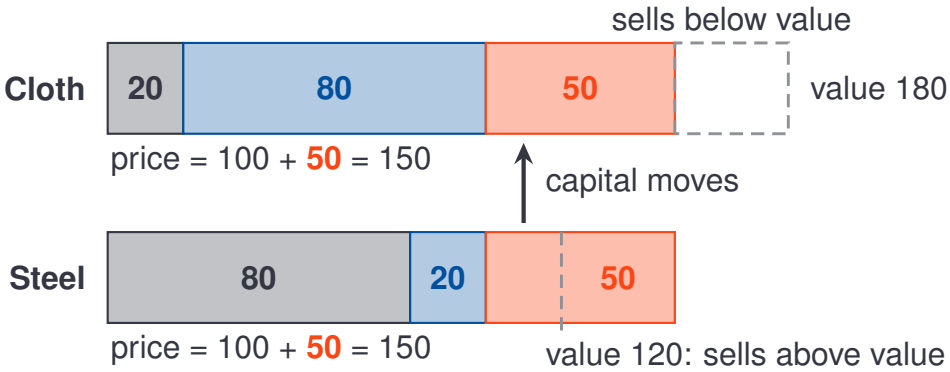


Capitalists as stockholders

Marx, *Capital*, vol. III, 1894

"So far as profits are concerned, the various capitalists are just so many stockholders in a stock company in which the shares of profit are uniformly divided per 100, so that profits differ [. . .] only in accordance with the amount of capital invested by each [. . .]."

Marx's answer: the totals

- Price of production: what it costs to make (**c** + **v**), plus profit at the average rate. Cloth and steel: 100 + **50** = 150 each.
- Cloth sells 30 below its value, steel 30 above: the deviations cancel.
- Total price 300 = total value 300. Total profit **100** = total surplus value **100**.

Marx (1894), vol. III, chs. 9–10.

▸ Marx's own table

Marx's laws of motion

Marx, *Capital*, vol. I, 1867

"[I]t is the ultimate aim of this work, to lay bare the economic law of motion of modern society"

Prediction	Where Marx says it
Workers' living standards	Manifesto (1848); *Capital* I, ch. 25
Wages vs. profits	*Wage Labour and Capital* (1849), ch. 8
The length of the working day	*Capital* I, ch. 10
The falling rate of profit	*Capital* III, chs. 13–14
Revolution in the most advanced countries	Manifesto; *Capital* I, ch. 32 and Preface

On Marx's theory of capitalism

- Labor power
- Surplus value
- The falling rate of profit

Böhm-Bawerk and the Close of Marx's System

Why does old wine cost more?

Old wine: same labor, higher price

- A cask of young wine sells for £10; after a dozen years in the cellar, £20.
- No labor went in during storage. Whose labor, then, was exploited?

Böhm-Bawerk (1884), pp. 364–365; Böhm-Bawerk (1889), p. 237.

Old wine: same labor, higher price

- A cask of young wine sells for £10; after a dozen years in the cellar, £20.
- No labor went in during storage. Whose labor, then, was exploited?
- Böhm-Bawerk: present goods are, as a rule, worth more than future goods; interest pays for the wait.

Böhm-Bawerk (1884), pp. 364–365; Böhm-Bawerk (1889), p. 237.

A note on Eugen von Böhm-Bawerk, 1851–1914

- Professor at Innsbruck (from 1881), later at Vienna (1904–14).
- Austrian finance minister three times: 1895, 1897–98, 1900–04.
- *Capital and Interest* (2 vols., 1884, 1889): why capital earns interest.
- *Karl Marx and the Close of His System* (1896).
- On Austria's 100-schilling note (dated 1984; in use 1985–2002).

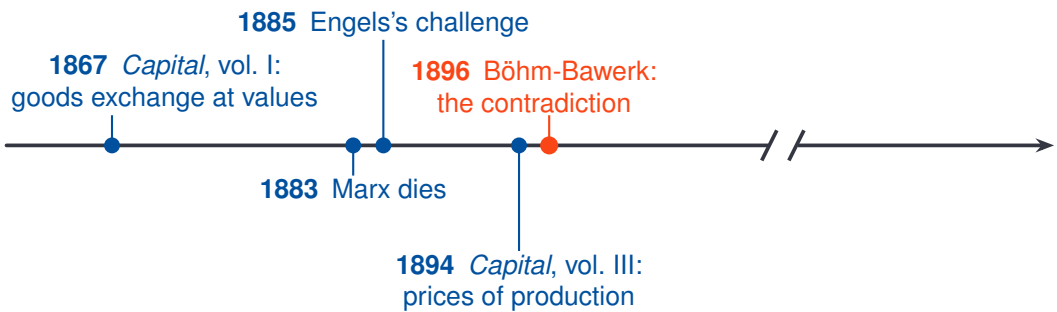
Böhm-Bawerk (1851–1914)

Zwiedineck-Südenhorst (1955).

From Volume I to Volume III

Law of value: goods exchange in proportion to the labor time needed to make them.

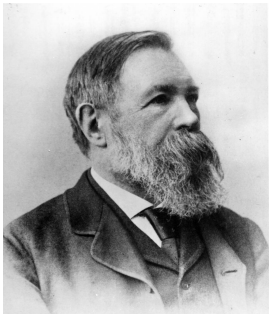
Transformation problem: getting from values (Volume I) to prices of production (Volume III).



Engels's challenge

Engels, Preface to *Capital*, Volume II, 1885

"If they can show in which way an equal average rate of profit can and must come about, not only without a violation of the law of value, but on the very basis of it, I am willing to discuss the matter further with them."



Friedrich Engels
(1820–1895)

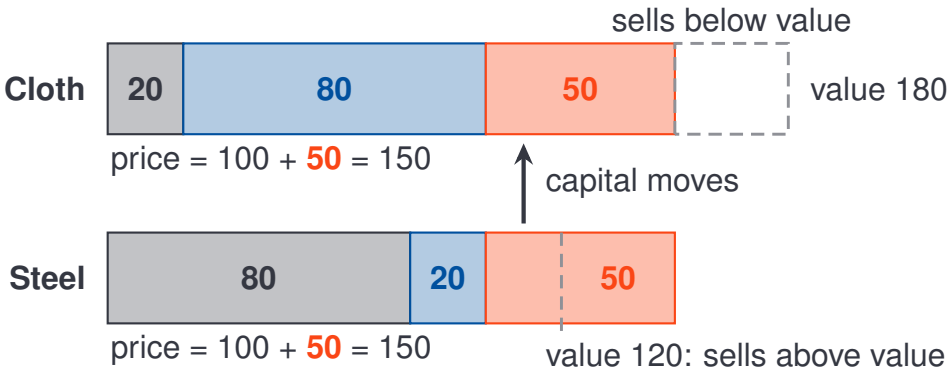
Böhm-Bawerk's verdict on Volume III

Böhm-Bawerk, *Karl Marx and the Close of His System*, 1896

"I cannot help myself; I see here no explanation and reconciliation of a contradiction, but the bare contradiction itself. Marx's third volume contradicts the first."

Böhm-Bawerk
(1851–1914)

The contradiction in numbers



- Volume I, at values: cloth : steel = 180 : 120.
- Volume III, at prices of production: 150 : 150. They cannot both hold.

Böhm-Bawerk on Marx's totals

Böhm-Bawerk, *Karl Marx and the Close of His System*, 1896

"[I]t is no answer to our question to be told the total price which they bring when taken altogether, any more than if, on asking by how many fewer minutes the winner in a prize race had covered the course than his competitor, we were to be told that all the competitors together had taken twenty-five minutes and thirteen seconds."

Elephants and mayflies

Böhm-Bawerk, *Karl Marx and the Close of His System*, 1896

"[E]lephants live on an average one hundred years and May flies only a single day, [. . .] we can strike an average of fifty years. [. . .] The deviations from this average 'mutually cancel each other,' and consequently on the whole and on the average the law that all kinds of animals have the same length of life is established!"

Marx's other defenses

Marx's defense	Böhm-Bawerk's reply
Labor governs the *movement* of prices	That makes labor one cause of prices, not the only one
Goods once exchanged at their values	No proof; and no producer is indifferent to waiting
Total surplus value fixes the average profit rate	Wages and the stock of capital fix it too

Böhm-Bawerk (1896), ch. 3, pp. 39–63.

A wage rise and prices of production

Cloth and steel again. Wages rise 10%; the same workers work the same hours.

▸ The wage rise, step by step

A wage rise and prices of production

Cloth and steel again. Wages rise 10%; the same workers work the same hours.

	Value (labor)	Price before	Price after
Cloth (labor-heavy)	180	150	154
Steel (machine-heavy)	120	150	146

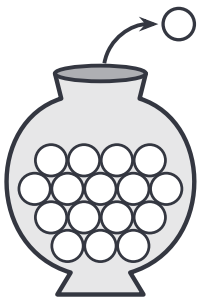
⇒ Böhm-Bawerk: the labor did not change, yet prices did. Wages help set prices.

▸ The wage rise, step by step

Böhm-Bawerk's urn

Böhm-Bawerk, *Karl Marx and the Close of His System*, 1896

"He acts as one who urgently desiring to bring a white ball out of an urn takes care to secure this result by putting in white balls only."



only white balls went in

white = made by labor

Böhm-Bawerk's urn

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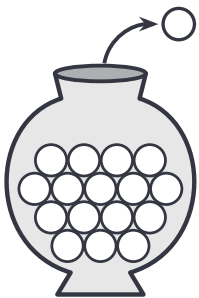

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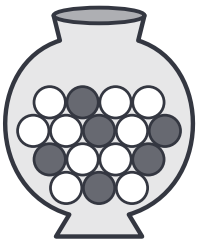


everything that is exchanged

dark = gifts of nature: soil, coal, water power

Is labor the only thing in common?

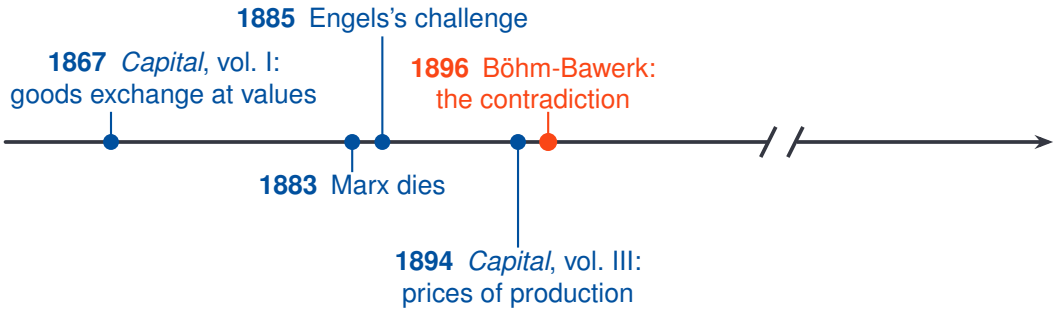
- Marx: goods traded as equals share a third thing, and it must be labor.

Is labor the only thing in common?

- Marx: goods traded as equals share a third thing, and it must be labor.
- Böhm-Bawerk: they also share scarcity, supply and demand, ownership, natural origin, cost.
- Marx's only argument for labor is negative (the common thing cannot be usefulness), and the same logic rules out labor.

Böhm-Bawerk (1896), ch. 4, pp. 68–76.

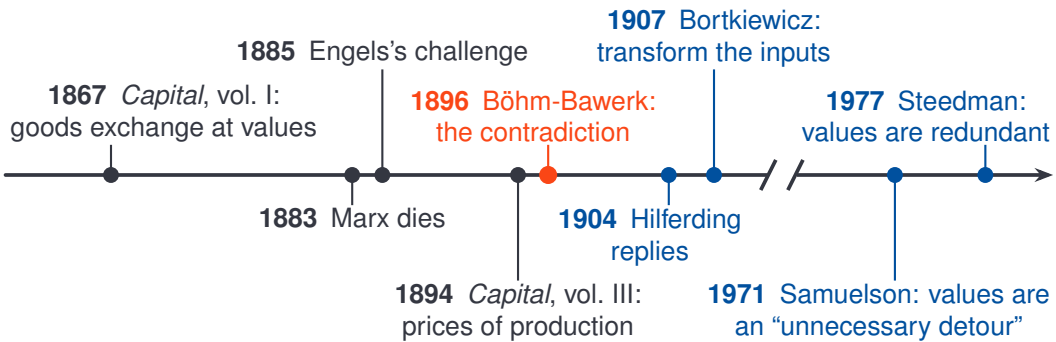
The transformation problem after Böhm-Bawerk



Duménil and Foley (2008).

▸ Bortkiewicz's correction

The transformation problem after Böhm-Bawerk



Duménil and Foley (2008).

▸ Bortkiewicz's correction

What survives of the labor theory of value?

- Labor is one cause of prices, not the only one:
  wages, capital and waiting move them too.
- Marx's question survives: where does profit come from?
- Böhm-Bawerk's answer is time, as in the wine:
  present goods are, as a rule, worth more than future goods.

Pareto’s Critique

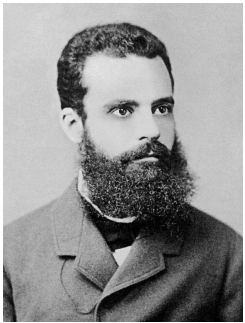
Capital vs. the capitalist

Pareto, Introduction to Marx's *Capital*, 1893

"Marx's book should be entitled The Capitalist *rather than* Capital*, at least if one takes the latter word in the fairly generally accepted sense of economic goods intended to facilitate the production of other goods."*

A note on Vilfredo Pareto, 1848–1923

- Born in Paris, 1848; civil-engineering degree, Turin, 1870.
- Railway engineer, then ironworks manager: general manager of the Società delle Ferriere Italiane, 1880–1890.
- Succeeded Walras in the chair of political economy at Lausanne, 1893.
- "Pareto's law" of incomes (*Cours*, 1896–97); "Pareto efficiency" (*Manuale*, 1906).
- Critic of Marx: Introduction to *Capital* (1893), *Les Systèmes socialistes* (1902–03), *Trattato di sociologia generale* (1916).



Vilfredo Pareto
(1848–1923)

Mornati (2014).

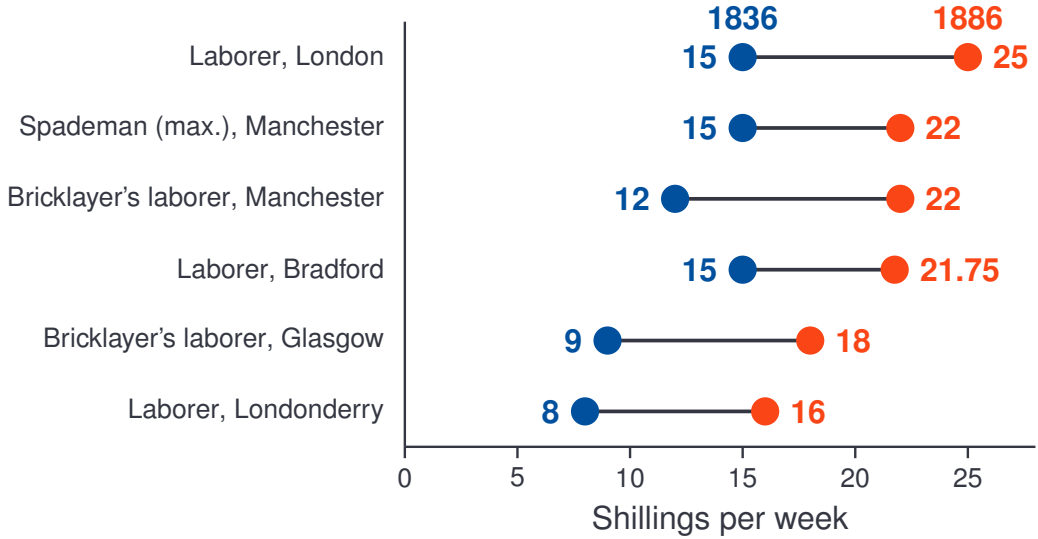
Who creates the surplus value?

The sewing-machine parody

- A seamstress rents a sewing machine for 30 centimes a day. Three hours of its work produce the rent and the 70 centimes she needs to live.
- She runs it six hours and keeps “the surplus value created by the machine.”
- “[A]nd that is what makes her laugh”: Pareto’s parody of *Capital* I, ch. 7.
  $\Rightarrow$ By Marx’s reasoning, any input can be cast as the exploited source of value.

Machine hours	What they produce
1–3	the machine’s rent + what she needs to live
4–6	“the surplus value created by the machine”

Pareto (1893), p. XXXIII.

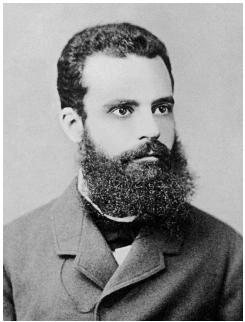


Note: money (not real) wages of unskilled non-farm labor, "fifty years ago" and "the present time" (about 1883–84), which Pareto labels 1836 and 1886. Consumer prices were roughly flat. Giffen (1886), p. 425, as printed in Pareto (1893), p. LXVI.

Pareto on the fall of aristocracies

Pareto, *Trattato di sociologia generale*, 1916

"Aristocracies do not last. Whatever the causes, it is an incontestable fact that after a certain length of time they pass away. History is a graveyard of aristocracies."



Vilfredo Pareto
(1848–1923)

Elites vs. classes

Elite: the few at the top who hold power, such as nobles, generals or party leaders.

- Pareto (1902): the class struggle is real, but it is fought among countless groups, “above all among the elites that contend for power.”
- A revolution does not end rule by a few: the winners become a new elite, and another rises against them.

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- Pareto (1902): the class struggle is real, but it is fought among countless groups, “above all among the elites that contend for power.”
- A revolution does not end rule by a few: the winners become a new elite, and another rises against them.
- Prediction for a socialist revolution: a new ruling class.

Pareto (1902), pp. 35–36, 117–118.

Djilas and the new class

Djilas, *The New Class*, 1957

"In contrast to earlier revolutions, the Communist revolution, conducted in the name of doing away with classes, has resulted in the most complete authority of any single new class. Everything else is sham and an illusion."

Milovan Djilas
(1911–1995)

Marx’s Predictions

Testing a conditional prediction

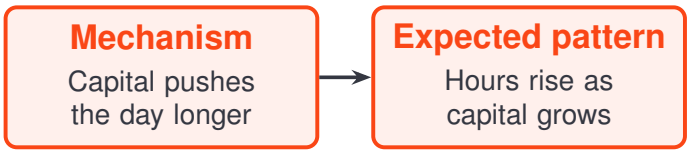
Conditional prediction: a force produces a pattern unless stronger forces push back.

Mechanism

Capital pushes the day longer

Testing a conditional prediction

Conditional prediction: a force produces a pattern unless stronger forces push back.



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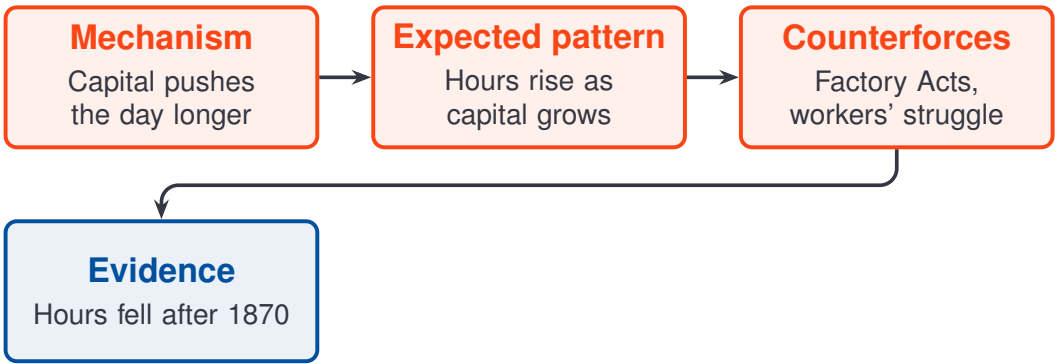

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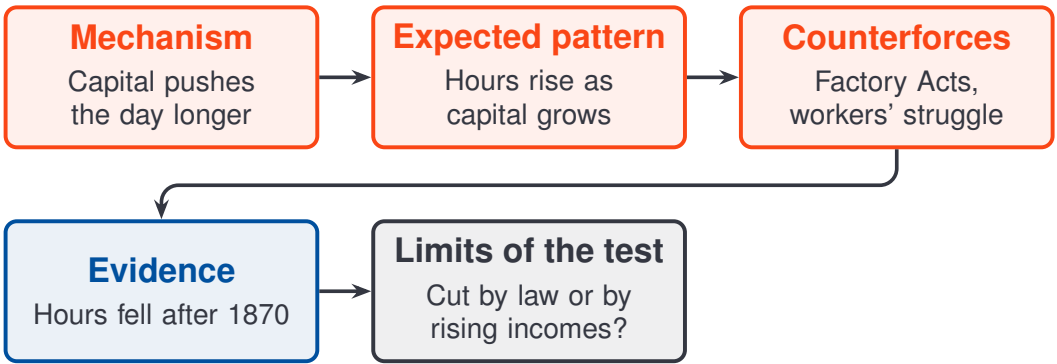
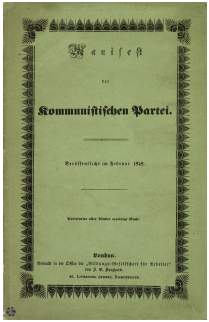
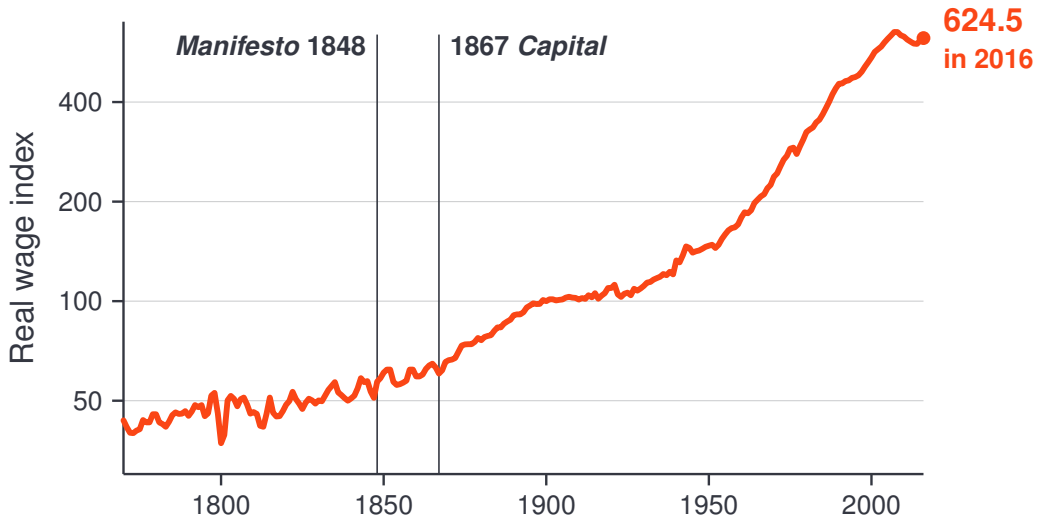


Marx and Engels, *Manifesto of the Communist Party*, 1848

"The modern labourer, on the contrary, instead of rising with the progress of industry, sinks deeper and deeper below the conditions of existence of his own class."


Manifest
der
Kommunistischen Partei.
London.

Manifest der Kommunistischen Partei
London, February 1848



Note: average real earnings in Britain, index 1900 = 100, log scale; Feinstein (1998) before 1880. Thomas and Dimsdale (2017).

Absolute vs. relative poverty

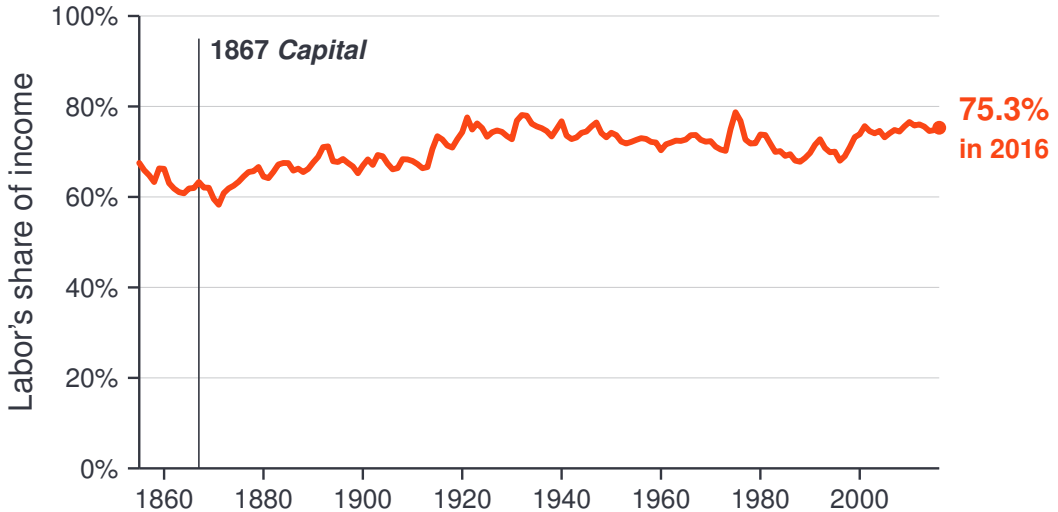
Marx, *Wage Labour and Capital*, 1849

"A house may be large or small; as long as the neighboring houses are likewise small, it satisfies all social requirement for a residence. But let there arise next to the little house a palace, and the little house shrinks to a hut."

Wages vs. profits

Marx, *Wage Labour and Capital*, 1849

"If capital grows rapidly, wages may rise, but the profit of capital rises disproportionately faster."



Note: labor's share of UK national income, excluding housing rents. Thomas and Dimsdale (2017).

Wages vs. profits: the record

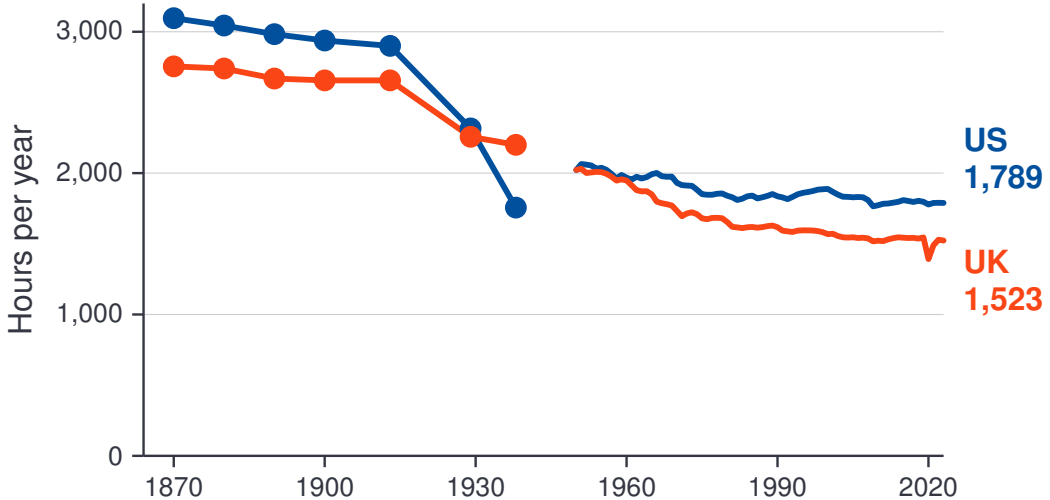
- Allen (2009): the profit rate doubled from 1800 to 1840, and capital's share rose from 20% (late 1700s) to over 40% (mid-1800s). Marx's era fits.
- Kaldor (1961), summing up the record to his day: in developed capitalist economies, a steady rate of profit and steady shares of profits and wages.
- Over 1975–2012 labor's share of corporate income worldwide fell about 5 points; 42 of 59 countries trended down (Karabarbounis and Neiman 2014).
- Limit: shares are measured in market prices, not in Marx's labor values.

Allen (2009); Kaldor (1961); Karabarbounis and Neiman (2014).

The length of the working day

- Marx: capital is “reckless of the health or length of life of the labourer, unless under compulsion from society”.
- The Factory Acts:
  - 1833: in textile mills (silk excepted), no children under 9; ages 9–13 at most 9 hours a day.
  - 1847: 10 hours a day for women and teenagers.
  - 1850: work only between 6 a.m. and 6 p.m., 10½ hours Monday to Friday, 7½ on Saturday: 60 hours a week.

Factory Acts (1833, 1847); Marx (1867), vol. I, ch. 10.



Note: annual hours worked per worker. 1870–1938: full-time production workers, benchmark years (dots), Huberman and Minns (2007); from 1950: all workers, Penn World Table 11.0. Via Our World in Data.

West (1983) on the working day

- Marx's theory needed the same rate of surplus value (e) in every industry, so the same working day everywhere.
- The inspectors' reports Marx cited: 181 mills worked men 11 to 15 hours a day, and a quarter of workers wanted 12.
- Better-paid workers tended to want shorter hours; half of the lowest-paid men wanted more than 10.

West (1983).

What would a falling rate of profit look like?

$$r = \frac{s}{c+v} = \frac{e}{1+q}$$

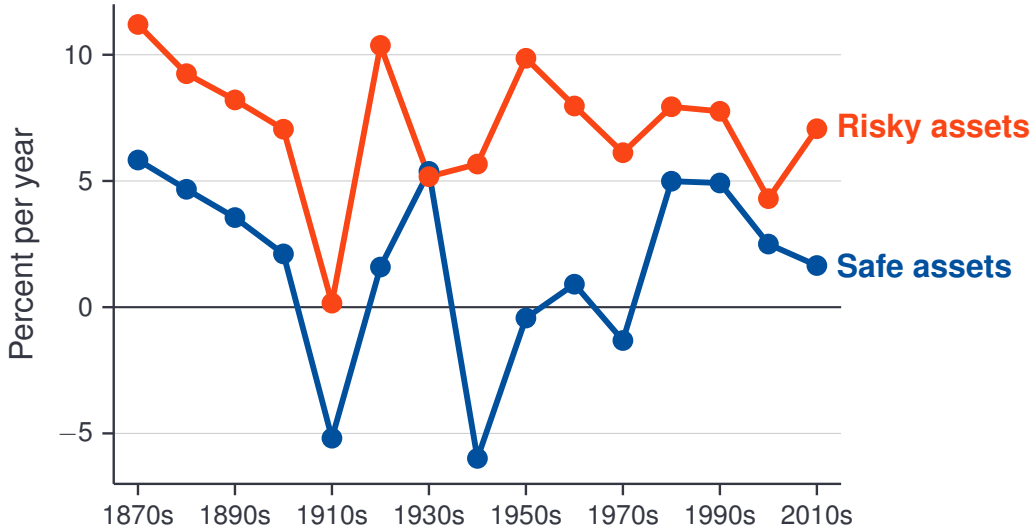
r: the rate of profit, profit per dollar advanced. $e = s/v$: the rate of surplus value.
$q = c/v$: the composition of capital, machines and materials per dollar of wages.

What would a falling rate of profit look like?

$$r = \frac{s}{c+v} = \frac{e}{1+q}$$

r: the rate of profit, profit per dollar advanced. $e = s/v$: the rate of surplus value.
$q = c/v$: the composition of capital, machines and materials per dollar of wages.

- Mechanism: competition pushes every mill to mechanize, so q climbs.
- Expected pattern: with e fixed at 100%, q rising from 1 to 4 cuts r from 50% to 20% (the 1830 and 1860 mills). A long downward drift.
- Counterforces: e rises, or machines get cheaper and q stops climbing.
- Evidence: the rate of return, what a dollar in shares and houses (risky) or government bonds (safe) earns in a year, after inflation.



Note: real returns (after inflation), decade averages across 16 rich countries weighted by GDP; before 1890 the risky series covers only two to five countries. Risky: equities and housing. Safe: government bonds and bills. Jordà et al. (2019).

Measuring Marx's rate of profit

- Marxian measures (profit per dollar of machines and buildings):
  - Maito (2014): the average profit rate trends down, in six rich countries from the 1850s and in eight others from the 1950s, to 2010.
  - Basu et al. (2025): the world profit rate fell 1960–2019, steeply to 1980, slowly after.
  - Basu and Manolakos (2013): weak evidence of a long-run fall in the U.S., 1948–2007.
- Market returns on shares and houses show little sign of a long downward drift (Jordà et al. 2019).
- Limits: all of these count capital and profit in market prices, not in Marx's labor values, and each study counts profits differently.

Maito (2014); Basu et al. (2025); Basu and Manolakos (2013).

Revolution, Empire and Slavery

Why did the revolution come in Russia?

Russia in 1917

Engels (1847): the revolution would come simultaneously in all civilized countries, at least England, America, France and Germany.

- 1897 census: about three in four workers were in agriculture (74.6%).
- Serfdom abolished in 1861; peasants got land in communal, not individual, ownership.
- Industry grew about 7% a year in the 1890s, faster than Germany or the U.S.
- The Putilov works in Petrograd employed more than 24,000 workers.

$\Rightarrow$ Not wholly feudal, but mostly peasant.

Engels (1847), Q. 19; Thorp and Thorp (1926), p. 235; Fischer (1994); Hasegawa (1981).

Marx and Engels on Russia

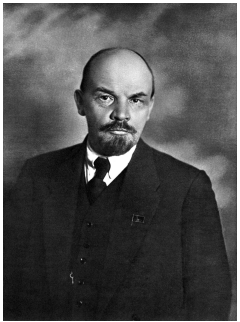
Marx and Engels, Preface to the Russian edition of the *Manifesto*, 1882

"If the Russian Revolution becomes the signal for a proletarian revolution in the West, so that both complement each other, the present Russian common ownership of land may serve as the starting point for a communist development."

Lenin's revisions of Marx

- *What Is to Be Done?* (1902): workers alone reach only "trade union consciousness"; a party of professional revolutionaries must lead.
- *Imperialism* (1916): "imperialism is the monopoly stage of capitalism"; capitalism survives by exporting capital and dividing the world.

⇒ Marx is not the Marxists.

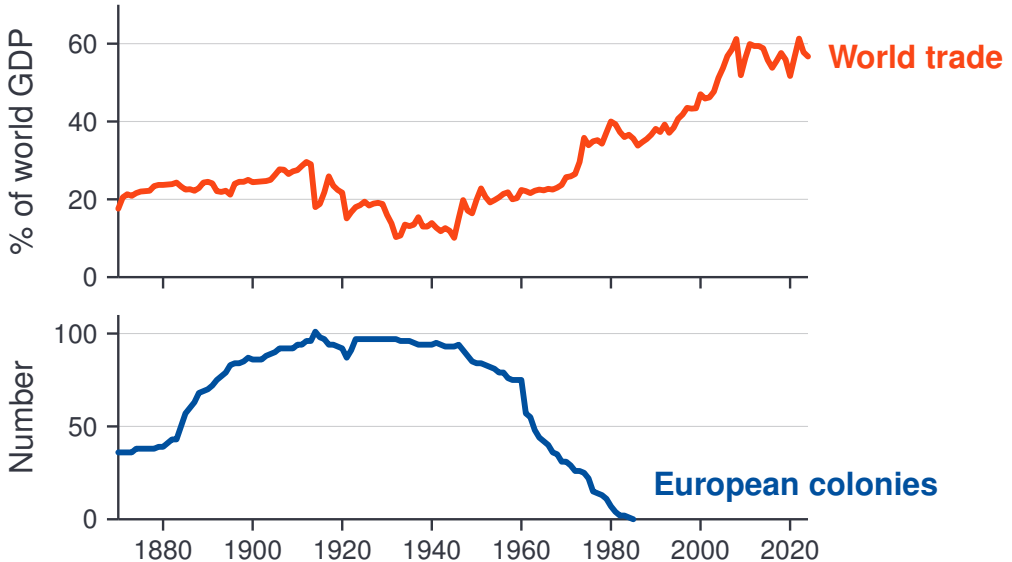


V. I. Lenin
(1870–1924)

Marx on British rule in India

Marx, “The Future Results of British Rule in India,” 1853

“England has to fulfill a double mission in India: one destructive, the other regenerating—the annihilation of old Asiatic society, and the laying the material foundations of Western society in Asia.”



Note: world exports plus imports; to 1949, Klasing and Milionis (2014), higher estimate; then Penn World Table and World Bank.
Colonies: today's countries then ruled by eight European powers (Becker 2023). Via Our World in Data.

Trade, empire and war

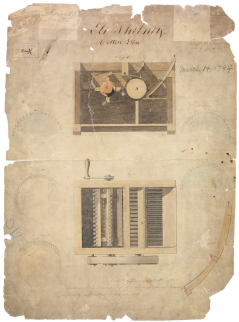
- Angell (1910): conquest does not pay. He did not say war was impossible, only futile.
- Davis and Huttenback (1986): the empire did not pay for Britain as a whole; some investors gained.
- Martin, Mayer and Thoenig (2008): countries that trade more with each other fight less; openness to world trade can raise the risk of war.

Angell (1910); Davis and Huttenback (1986), p. 306; Martin et al. (2008).

Slavery and the cotton market

Marx, *Capital*, vol. I, 1867

"[A]s soon as people, whose production still moves within the lower forms of slave-labour [. . .] are drawn into the whirlpool of an international market [. . .] the civilized horrors of over-work are grafted on the barbaric horrors of slavery, serfdom, &c."



Whitney's cotton gin patent, 1794

Did slavery need new fertile land?

Marx, “The North American Civil War,” 1861

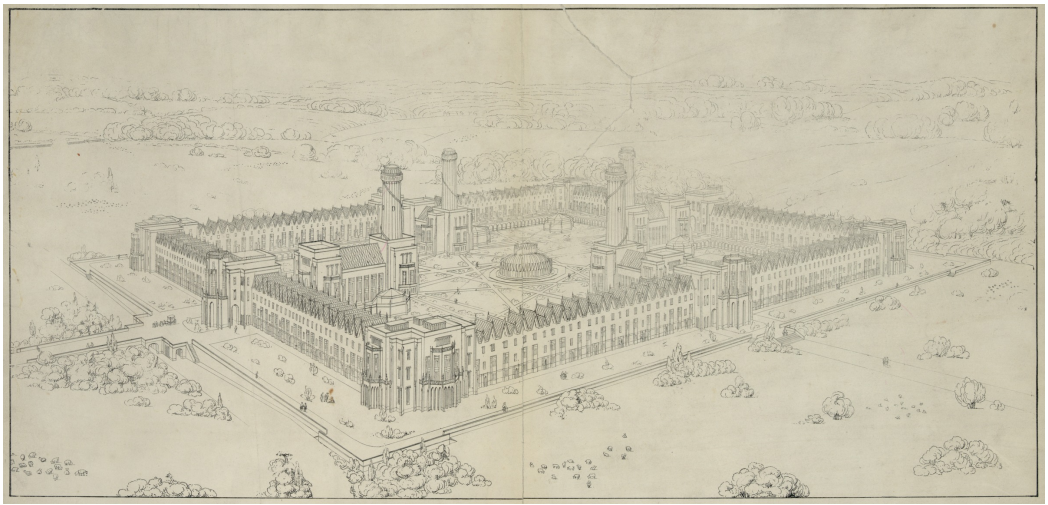
“The cultivation of the southern export articles, cotton, tobacco, sugar, etc., carried on by slaves, is only remunerative as long as it is conducted with large gangs of slaves, on a mass scale and on wide expanses of a naturally fertile soil, which requires only simple labour.”

Slavery: what the record shows

- Olmstead and Rhode: cotton picked per slave per day rose from 28 lb (1801) to 114 lb (1862), mainly through new cotton varieties.
  $\Rightarrow$ More cotton per slave, not only more land.
- Abolition: Britain 1833 (in force 1834), France 1848, U.S. 1865, Cuba 1886, Brazil 1888.
- Enslaved people were about 15–20% of U.S. wealth in 1860, and 40–50% of Southern wealth.

Olmstead and Rhode (2008); Dray et al. (2026), pp. 27–28; Wright (2006), p. 60.

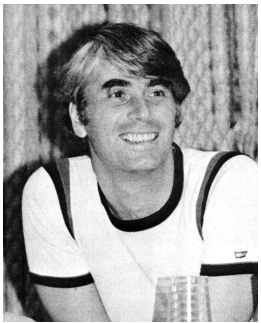
Was Owen free to try his utopia in America?



Nozick on socialism and freedom

Nozick, *Anarchy, State, and Utopia*, 1974

"The socialist society would have to forbid capitalist acts between consenting adults."



Robert Nozick
(1938–2002)

Marx's predictions: the record

Prediction	The record	Verdict
Workers sink deeper into poverty	UK real wages up about elevenfold, 1848–2016	Wrong: wages lagged, then rose
Profits outgrow wages	Profit rate doubled 1800–40; shares steady later	Right for 1800–40, then mixed
A longer working day	Annual hours per UK worker: 2,755 (1870), 1,523 (2023)	Wrong: hours fell as incomes rose
A falling rate of profit	Little fall in market returns; Marxian measures fall	Mixed: depends on the measure
Revolution in the most advanced countries	In mostly peasant Russia, 1917	Wrong; the 1882 Russian path was conditional
Capitalism needs empire (Lenin)	Colonies vanished; trade 47% of GDP by 2000	Wrong, and Lenin's claim, not Marx's

On Marx and his critics

- The transformation problem
- Conditional predictions vs. the record
- Marx is not the Marxists

Marx's own example: five spheres

Marx, *Capital* III, ch. 9: every capital = 100, e = 100%, all **c** used up.

Sphere	c + v	s	Value	Price	Deviation
I	80c + 20v	20	120	122	+2
II	70c + 30v	30	130	122	−8
III	60c + 40v	40	140	122	−18
IV	85c + 15v	15	115	122	+7
V	95c + 5v	5	105	122	+17
Total	**500**	**110**	**610**	**610**	**0**

- Average rate of profit: **110** / 500 = 22%, so each capital of 100 earns 22.
- Marx: each fifth of the product sells at 122. Deviations cancel: +26 and −26.

Back

Bortkiewicz's correction

Department	At values: c + v + s	At prices: c + v + profit
I. Means of production	225 + 90 + 60 = 375	288 + 96 + 96 = 480
II. Workers' goods	100 + 120 + 80 = 300	128 + 128 + 64 = 320
III. Capitalists' goods (gold)	50 + 90 + 60 = 200	64 + 96 + 40 = 200
Total	375 + 300 + 200 = **875**	480 + 320 + 200 = **1,000**

Price the inputs too, and here only one total survives: profit = surplus value (200), but price $\neq$ value (1,000 vs. 875).

Bortkiewicz (1907), Tables 1–2, p. 204.

Back

The wage rise, step by step

	c	v	s	Value	Capital	Price
Cloth	20	80 → 88	80 → 72	180	108	154.29
Steel	80	20 → 22	20 → 18	120	102	145.71

1. Same workers, same hours: the new value each industry creates (**v** + **s**) stays 160 in cloth and 40 in steel. Wages rise 10%, so **v** takes more of it and **s** shrinks.
2. Average rate of profit: **90** of surplus value on a capital (**c** + **v**) of 108 + 102 = 210, so $r = 3/7 \approx 42.9\%$.
3. Price = capital $\times (1 + 3/7)$ = capital $\times 10/7$: cloth 154.29, steel 145.71, total 300. Values stay 180 and 120; before the rise, both prices were 150.

Back

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